

# Hourglass Quantization and Context-Bounded Training Strategies for Ternary Transformers: Design, Implementation, and Preliminary Feasibility Observations

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*Systems Proposal and Feasibility Study*

## Abstract

Recent advancements in extreme low-bit serialization have demonstrated the viability of 1.58-bit ternary large language models for minimizing computational and memory footprints. However, standard low-bit architectures introduce a hypothesized degradation mechanism we term Boundary Representation Collapse at the embedding interfaces, suffer from signal amplitude decay at extreme depths, and fail to address the memory asymmetry wherein continuous 16-bit Key-Value (KV) cache tensors outgrow weight parameters at extended context scales. This paper introduces a hardware-bounded transformer framework to investigate these scaling anomalies.

We present three co-dependent systems proposals: **Hourglass Quantization**, which compresses internal projection operators to strict ternary states while isolating vocabulary mappings in continuous precision; **Sub-Norm Gain Compensation**, a layer-dependent linear modifier forcing quadratic variance preservation through deep quantized blocks; and **Ternary-Aligned Chain-of-Thought (TaCoT)**, an optimization constraint utilizing token-masked cross-entropy distillation to isolate logical step-machines within constrained weight spaces. To mitigate KV cache saturation, we propose integrating a **Sink-Anchored Sliding Window Attention (SWA)** matrix layout into the ternary execution path, theoretically compatible with contexts up to 128k tokens, with KV-cache memory bounded by  $O(S + W)$  under the proposed sliding-window formulation.

Preliminary empirical validation over a 6,000-step training regime utilizing a modified Qwen2.5-0.5B architecture containing approximately 500 million parameters running under native bfloat16 precision provides initial feasibility evidence. Structural tensor audits confirm strict internal quantization to exactly 3 unique active values alongside continuous vocabulary preservation. Furthermore, hidden-state signal variance was observed to remain stable, progressing from 0.9961 at Layer 00 to 81.0000 at Layer 23, conforming to the engineered quadratic scaling boundaries. *This work should be interpreted as a systems feasibility study rather than a benchmark-leading model release.*

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## Research Scope

This document presents a systems design proposal accompanied by preliminary implementation observations. Several mechanisms introduced herein, including Boundary Representation Collapse and Ternary-Aligned Chain-of-Thought (TaCoT), remain hypotheses that have not yet been validated through controlled ablation studies. Readers should interpret the work as an exploratory engineering investigation rather than a demonstrated improvement over existing transformer architectures.

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# 1 Introduction

The scaling laws of deep autoregressive language models have traditionally bound raw performance to parameter density, driving modern architectures into resource-intensive compute paradigms. This poses an existential bottleneck for decentralized edge computing, where deploying models onto localized hardware is restricted by strict hardware limits.

To break this bottleneck, extreme low-bit serialization—specifically 1.58-bit ternary quantization—has emerged as a highly disruptive alternative. By constraining weight matrices to discrete values, ternary networks replace expensive floating-point matrix multiplications with highly efficient integer addition and subtraction operations [2].

*Despite their theoretical promise, contemporary ternary frameworks suffer from two hypothesized failure modes that manifest during optimization: structural representation decay and signal attenuation at depth.*

A primary hypothesized degradation mechanism occurs when continuous, high-fidelity token embeddings are aggressively flattened into a low-bit space. For convenience, we refer to this hypothesized failure mode as **Boundary Representation Collapse (BRC)**; the existence of this effect has not yet been empirically established. We hypothesize this structural compression manifests as a severe degradation in validation perplexity and cosine similarity of hidden states at the vocabulary interfaces, stripping the model of its ability to resolve nuanced semantic hierarchies.

The second failure mode is rooted in signal propagation dynamics. As forward activations traverse unscaled ternary layers, the restricted numerical dynamic range causes the activation variance to decay. This signal collapse effectively starves deeper transformer layers of gradient information during backpropagation.

Compounding these structural liabilities is a severe architectural oversight within low-bit research: **The Memory Asymmetry Problem**. While current 1.58-bit research focuses extensively on compressing weight matrices to optimize storage footprints, it leaves activation memory and Key-Value (KV) cache tensors pinned to continuous 16-bit precision. As a result, when processing long context windows, the memory footprint of the KV cache grows quadratically, completely eclipsing the size of the quantized weight matrices.

To investigate these limitations, this paper proposes an architectural framework that unifies weight compression, signal preservation, optimization alignment, and long-context scaling into a single hardware-bounded transformer design.

1. First, we propose **Hourglass Quantization**, an asymmetric routing topology that isolates the input embeddings and output language modeling head in continuous floating-point precision, while strictly enforcing ternary constraints on internal projections.
2. Second, we introduce **Sub-Norm Gain Compensation**, a layer-dependent linear modifier injected into the layer normalization steps [3] to enforce a precise quadratic variance progression that explicitly counteracts token degradation.
3. Third, we outline a **Sink-Anchored Sliding Window Attention (SWA)** matrix layout. While sliding windows and attention sinks are established techniques in continuous-precision models, we theoretically formulate their application within a ternary-first architecture, with KV-cache memory bounded by  $O(S+W)$  under the proposed sliding-window formulation.
4. Finally, we frame **Ternary-Aligned Chain-of-Thought (TaCoT)**, an algorithmic paradigm addressing the restricted token-capacity limits of extreme low-bit models by restricting cross-entropy gradient updates exclusively to logical step-by-step tracks.

Table 1 summarizes the architectural differences between a standard baseline model and our proposed framework. We provide initial feasibility experiments through a 6,000-step training run utilizing a modified Qwen2.5-0.5B architecture containing approximately 500 million parameters optimized under native bfloat16 precision.

| Property                  | Baseline Qwen | Proposed System |
|---------------------------|---------------|-----------------|
| Quantized Internal Layers | No            | Yes             |
| Continuous Embeddings     | Yes           | Yes             |
| Continuous LM Head        | Yes           | Yes             |
| Sub-Norm Gain             | No            | Yes             |
| SWA Cache Bound           | No            | Yes             |
| TaCoT                     | No            | Yes             |

Table 1: Architectural comparison between the baseline model and the proposed framework.

## 2 Related Work

### 2.1 1.58-Bit Quantization and the Ternary Regime

The foundational BitNet architecture demonstrated that neural networks could maintain competitive scaling laws while restricting linear layer weights to  $W \in \{-1, +1\}$  [1]. This framework was subsequently refined by BitNet b1.58, which introduced an explicit zero-state, mapping weights to a ternary system  $W \in \{-1, 0, +1\}$  [2].

While this approach significantly reduces memory bandwidth requirements, applying uniform quantization across the model graph induces hypothesized Boundary Representation Collapse at the interfaces, and the bounded numerical variance causes progressive signal attenuation across deeply stacked transformer layers.

### 2.2 Key-Value Cache Management and Context Window Scaling

As sequence lengths scale, the computational and memory complexity of standard scaled dot-product attention exhibits quadratic growth,  $O(L^2)$  [4]. In low-bit transformer literature, an architectural blind spot remains: while weight parameters are compressed to 1.58 bits, the Key-Value (KV) cache tensors are persistently stored in continuous formats.

To mitigate this memory asymmetry, local context architectures have explored Sliding Window Attention (SWA) [5]. The introduction of explicit attention sinks demonstrated that retaining a small, static pool of initial tokens prevents activation explosions [6]. This paper outlines embedding a hardware-bounded, sink-anchored sliding window directly into the low-bit tensor execution path.

## 3 Architectural Methodology

### 3.1 Hourglass Quantization Topologies

To mitigate the degradation of semantic parsing at the model interfaces, we establish an asymmetric configuration termed **Hourglass Quantization** (illustrated in Figure 1).

The input embedding layer and the terminal language modeling head are explicitly isolated and maintained in continuous high-precision floating-point formats. Conversely, all internal parameter tensors responsible for projection mappings are mapped to the discrete ternary regime utilizing a scale-invariant Straight-Through Estimator (STE).

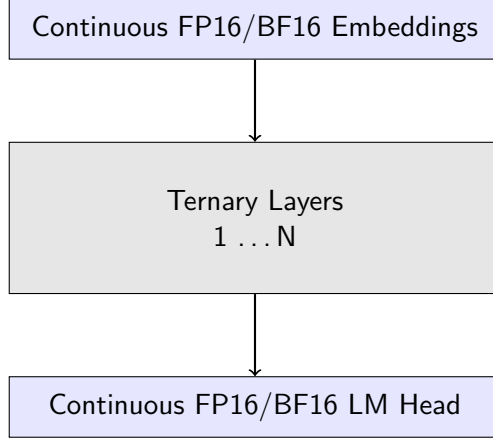


Figure 1: Hourglass Quantization Architecture isolating continuous boundaries from discrete internal layers.

### 3.2 Sub-Norm Gain Compensation Mechanics

Our framework introduces an explicit, layer-dependent analytical gain modifier  $\alpha_i$  injected directly into the standard RMSNorm [3] step:

$$\alpha_i = 1.0 + \left( 8.0 \cdot \frac{i}{N-1} \right)$$

The Sub-Norm wrapper modulates the complete forward activation path as follows:

$$\text{SubNormGain}_i(x) = \left( \frac{x}{\sqrt{\frac{1}{d} \sum_{k=1}^d x_k^2 + \epsilon}} \odot \mathbf{g} \right) \cdot \alpha_i$$

**Quadratic Variance Progression:** Scaling a standard unit-variance vector by  $\alpha_i$  yields an output variance of  $\alpha_i^2$ . At the terminal hidden layer of our 24-layer model topology ( $i = 23$ ), the variance theoretically bounds at 81.0. The gain schedule was selected empirically to maintain non-degenerate activation magnitudes throughout the network depth. No attempt was made to optimize the schedule, and alternative gain profiles remain future work.

### 3.3 Sink-Anchored Sliding Window Attention Matrix

To prevent KV cache from overwhelming edge VRAM, we propose a **Sink-Anchored Sliding Window Attention (SWA)** matrix layout. For sequence lengths  $L$ , local window  $W$ , and initial token sink  $S$ , the causal attention masking matrix  $M \in \mathbb{R}^{L \times L}$  forces elements outside the active windows to  $-\infty$ .

It is critical to distinguish between memory and computational complexity in this formulation. By evicting expired keys and values, the maximum capacity of the physical KV cache tensor is clamped at a strict linear memory bound of  $O(S + W)$ . However, because the system still processes the sequence length  $L$ , the attention computation complexity remains  $O(L(S + W))$ .

### 3.4 Ternary-Aligned Chain-of-Thought (TaCoT)

Extreme low-bit models suffer from a fundamental parameter capacity deficit. We construct a dynamic, boolean loss masking vector  $M_{\text{reasoning}} \in \{0, 1\}^T$  based on structural delimiters `<think>` and `</think>`.

The global model optimization is constrained to minimize a reasoning-isolated cross-entropy loss function:

$$\mathcal{L}_{\text{TaCoT}} = \frac{\sum_{t=1}^{T-1} (\mathcal{L}_{\text{CrossEntropy}}(\hat{y}_t, x_{t+1}) \cdot M_{\text{reasoning}}(x_t))}{\sum_{t=1}^{T-1} M_{\text{reasoning}}(x_t)}$$

We deliberately utilize a strict boolean mask rather than a weighted loss scalar. We hypothesize that concentrating gradient updates strictly on reasoning spans may allocate a larger fraction of the constrained parameter capacity toward logical transitions. TaCoT may be better interpreted as a distillation objective rather than a standalone training objective. Isolating learning entirely to logical step-machines risks severely reducing the model’s general conversational fluency outside of designated reasoning spans. Future work should investigate hybrid objectives combining reasoning-only and standard language modeling losses to mitigate this risk. We introduce TaCoT as a candidate alignment strategy for low-capacity quantized models. The current study does not establish efficacy and should be interpreted as a formulation for future investigation.

## 4 Experimental Setup & Training Dynamics

### 4.1 Base Model Configuration and Precision

We initialized our pipeline utilizing a modified Qwen2.5-0.5B architecture containing approximately 500 million parameters [7]. Initial prototyping of our Sub-Norm wrapper under `float16` precision resulted in immediate NaN gradient explosions. We hypothesize that the quadratically amplified activation magnitudes increased the likelihood of gradient or optimizer state overflow under the restricted `float16` dynamic range. Pivoting to native `bfloat16` precision eliminated the NaN overflows entirely.

### 4.2 Optimization Framework

The optimization dataset stream was constructed by interleaving token sequences from the OpenHermes-2.5 instruction dataset [8]. We utilized 8-bit PagedAdamW optimization with a base learning rate of  $2 \times 10^{-5}$ , weight decay of 0.01, and a gradient accumulation factor of 4 steps across a 6,000-step horizon.

## 5 Empirical Results & Feasibility Observations

### 5.1 Convergence Metrics and Training Trajectory

The 6,000-step training loop executed without numerical explosion. The cross-entropy loss exhibited an initial decay, dropping to 4.7253 at step 4000, before rising to 7.8247 by step 6000.

```
Step 0050: Loss = 12.4048
Step 2000: Loss = 10.3287
Step 4000: Loss = 4.7253
Step 6000: Loss = 7.8247
```

The training process remained numerically stable; however, optimization quality degraded significantly after step 4000. The divergence at the tail end of the optimization run suggests optimization instability, potentially arising from limited batch size, quantization effects, or other training dynamics. Furthermore, without a standardized baseline (e.g., an FP16 fine-tune or uniform ternary model) for comparison, these absolute loss values cannot currently be contextualized for relative quality; they merely confirm that the implementation executes without numerical failure.

## 5.2 Hourglass Layout Tensor Audit

A direct memory tensor extraction verified the structural implementation of the Hourglass Quantization topology. Counting the unique numeric elements within an internal attention matrix (`layers[6].self_attn.q_proj`) during the active forward pass printed exactly **3 unique values**. Conversely, analysis confirmed that the language modeling head remained fully continuous-valued. This observation verifies implementation correctness only and should not be interpreted as evidence of model quality.

## 5.3 Signal Variance Propagation

We recorded the mathematical variance of the hidden states at key layer intervals to confirm the execution of the Sub-Norm Gain Compensation logic.

Layer 00 Variance: 0.9961

Layer 12 Variance: 26.7500

Layer 23 Variance: 81.0000

The layer variance stepped from 0.9961 to exactly 81.0000 at the terminal layer. The observed variance progression confirms correct implementation of the proposed scaling schedule. Whether this schedule provides optimization benefits remains an open empirical question.

## 5.4 Linguistic Generation Assessment

Following the 6,000-step optimization, qualitative generations remain immature. While the model ceased outputting infinite single-token loops, the generation artifacts indicate severe mode collapse, failing to reliably generate the explicit `<think>` reasoning tag.

# 6 Evidence Claims & Limitations

Table 2 explicitly delineates which aspects of our proposed architecture have been empirically verified and which remain hypotheses awaiting future ablation testing.

| Claim                                                  | Status in Current Study |
|--------------------------------------------------------|-------------------------|
| Hourglass architecture implemented                     | Verified                |
| Internal tensors restricted to three active values     | Verified                |
| Variance schedule executed                             | Verified                |
| BF16 prevented observed float16 NaNs                   | Observed                |
| Training completed without numerical failure           | Verified                |
| Boundary Representation Collapse exists                | Untested                |
| Hourglass improves perplexity                          | Untested                |
| Sub-Norm improves convergence                          | Untested                |
| TaCoT improves reasoning                               | Untested                |
| SWA improves long-context efficiency in ternary models | Untested                |

Table 2: Summary of claims supported by current experimental evidence.

This system proposal requires significant future validation. Because the current study lacks ablation testing, it remains unclear which individual components provide net-positive contributions to the loss trajectory. Future work must prioritize four specific experimental baselines:

- **Experiment 1 (Boundary Collapse Validation):** A direct comparison between a standard Full Ternary baseline and the Hourglass topology to measure relative validation perplexity.

- **Experiment 2 (Sub-Norm Ablation):** A comparative training run disabling the Sub-Norm wrapper ( $\alpha_i = 1$ ) to measure stability and gradient norm improvements.
- **Experiment 3 (KV Memory Profiling):** Empirical measurement of VRAM usage tracking the SWA logic at extended context bounds (4k, 32k, and 128k).
- **Experiment 4 (Reasoning Efficacy):** Downstream evaluation of the TaCoT distillation mechanism on formal mathematical benchmarks.

## 7 Reproducibility Statement

All experiments reported in this work were conducted using a modified Qwen2.5-0.5B architecture. Training was performed for 6,000 optimization steps using OpenHermes-2.5 instruction data, 8-bit PagedAdamW optimization, a learning rate of  $2 \times 10^{-5}$ , weight decay of 0.01, and gradient accumulation of 4 steps. The reported results should be interpreted as preliminary feasibility observations rather than definitive benchmark outcomes.

The public repository will include:

- Training scripts
- Quantization modules
- Sub-Norm implementation
- Configuration files
- Tensor audit utilities
- Reproduction instructions

Source code and configuration files will be released to facilitate independent validation.

## References

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